

**Advancing towards clinical process support:
Erasing the pain of a low back pain
management pathway**

OpenEHR Fellowship 2025-2026

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This report documents a fellowship exploring clinical process support through a proof of concept that models the low back pain management pathway using openEHR for clinical data representation and BPMN/DMN for process orchestration. It traces the architectural decisions, trade-offs, and open questions that emerge when computable process support is applied to a real condition in a real health system.

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Preface

This document is the final report of a fellowship undertaken through openEHR International. It concerns a question I have been circling for several years from a clinical and health informatics perspective: why do we build sophisticated systems to capture what clinicians do, and then leave them largely alone to figure out what to do next?

Seems easy but it is extremely complex. The data is there, increasingly structured and increasingly computable. The clinical guidelines are there, with decades of evidence behind them. The gap between the two is not one of information but of integration — of a layer that takes structured clinical data and converts it into guidance at the moment of care. This fellowship was an attempt to understand that layer: what it is made of, which tools can build it, and what happens when you try to apply those tools to a real clinical condition in a real health system.

Low back pain was the chosen test case, for reasons the first two sections explain in detail. The more important admission is that the choice also reflected pragmatism: the condition has a reasonably well-specified pathway, a clinical team willing to engage, and a high enough volume in Catalonia to make the investment worthwhile if the architecture proves replicable. A fellowship is a structured experiment. This one produced a proof of concept, a set of architectural decisions, and a clearer picture of where the field needs to go.

The document is structured as a decision log as much as a technical report. Each section surfaces trade-offs, makes choices explicit, and tries to leave useful breadcrumbs for anyone who attempts something similar. It does not claim to have solved the problem of clinical process support; it claims to have navigated a corner of it honestly.

All technical artifacts can be found in the github repository: https://github.com/xabimich/openehr_fellowship_lbp

Acknowledgements

This work would not have been possible without the openEHR International fellowship programme, which provided the structure, the community, and the intellectual context. Thanks to Rong Chen, Seref Arıkan, Thomas Beale and Rachel Dunscombe for their support.

The clinical content draws on the low back pain pathway developed by the ICS (Institut Català de la Salut). I am grateful to the clinical team there for their willingness to engage with a modelling exercise that asked them to think about their own practice in ways that clinical informatics rarely makes visible. Thanks to Adriana Burítica, Maria José Dura Mata and Francisco Javier Medel.

Technological support was provided through the Open Platform project at CTTI, in collaboration with contract adjudicators NTT Data and Tietoevry. Special thanks to Ingrid Soler, Anna Navarro, Sylvie Vergé, and Jesús Pulgarín.

Finally, I would like to thank my colleagues at CatSalut, particularly those most directly involved: Andrea Climent, Lluís Valle, Jordi Piera, and David Alonso.

Xabier Michelena

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Setting the Challenge

Every clinician has felt it: the weight of a system designed to register rather than guide. Electronic health records have become the backbone of modern healthcare yet for all the investment poured into their adoption, they remain, at their core, sophisticated filing cabinets. They capture what happened (for it to potentially be reimbursed). They rarely help you decide what should happen next.

This tension between EHRs as documentation tools and EHRs as process support tools is the starting point for this fellowship. And low back pain is where we chose to test whether the gap can be closed.

The documentation trap

The promise of the electronic health record was transformative: a unified, computable representation of the patient's health story, accessible across settings and over time. In many ways, that promise has been kept. Structured data entry, terminology bindings, interoperability profiles, and standards like openEHR and HL7 FHIR have made clinical data more portable and reusable than ever before.

But the clinical process the orchestrated sequence of assessments, referrals, decisions, and follow-ups that constitutes care has largely been left outside this revolution. Pathways are documented in PDFs. Guidelines live in portals. Decision support, where it exists at all, is grafted onto systems as alerts rather than woven into workflows. The result is that clinicians navigate a persistent gap between knowing what the data says and knowing what to do with it.

Once you begin thinking seriously about building digital tools that support a clinical process, you quickly discover that data persistence is necessary but not sufficient. You need the data to be structured (openEHR handles this well), you need it to be interoperable (FHIR helps bridge that), and you need something to orchestrate what happens with it a layer that has no settled name and no settled home in most health system architectures. Seref Arikan's concept of the *EHR Slice* captures part of this ambition: a packaged unit that bundles FHIR API definitions, openEHR archetypes, templates, and technology-agnostic mappings into something closer to a deployable clinical microservice ([Arikan 2024](#)). I will try to humbly add another layer: an explicit, computable representation of what the clinical workflow is meant to do.

This fellowship is an attempt to explore that third layer: clinical process standardisation.

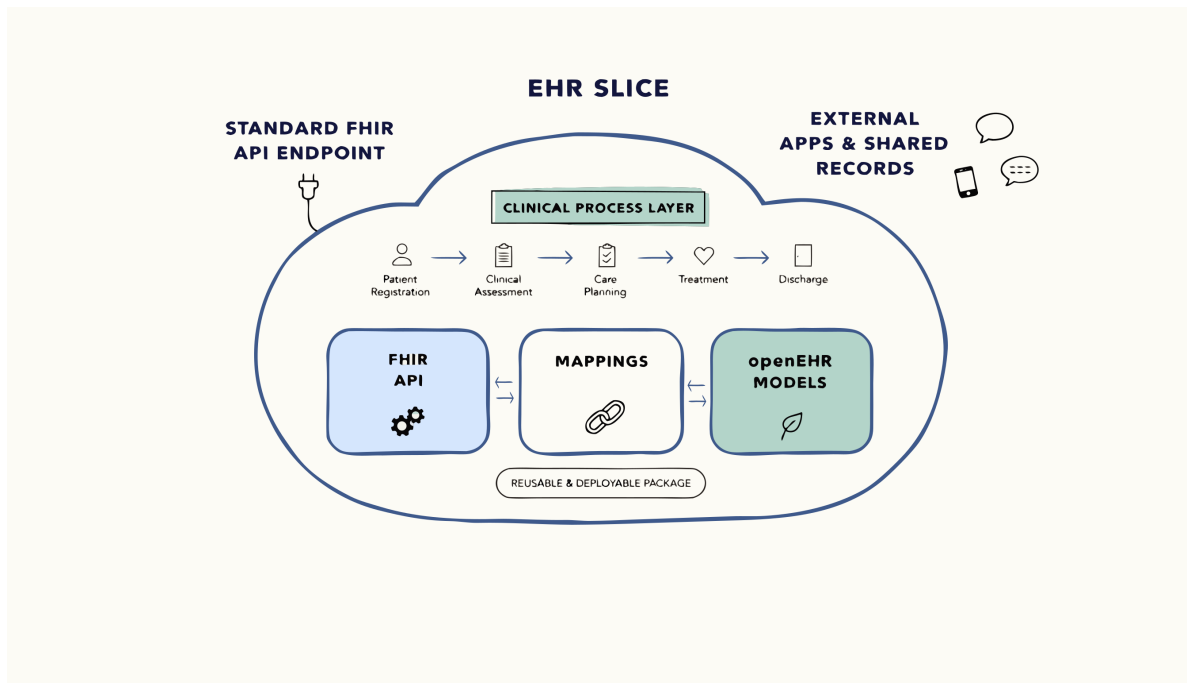


Figure 1: The EHR Slice concept: a deployable unit bundling FHIR API definitions, openEHR archetypes and templates, and mappings — the foundation on which this fellowship builds a process orchestration layer.

Why low back pain?

Low back pain is not a glamorous condition which is precisely what makes it a sharp test case. It is the leading cause of years lived with disability worldwide, affecting an estimated 619 million people globally in 2020, a figure projected to rise further as populations age ([Ferreira et al. 2023](#)). In Catalonia, as in most health systems, it is one of the highest-volume conditions managed across both primary and secondary care, with care fragmented across physiotherapy, primary care, rheumatology, pain units, and imaging services.

It also arrives with something rare: a relatively well-defined clinical pathway. Clear criteria separate non-specific from specific causes, acute from chronic presentations, and patients who need reassurance from those who need imaging, specialist review, or structured rehabilitation. International guidelines converge on a broadly similar sequence of decisions. That structure makes it modelable. The pathway crosses organisational levels, involves multiple professions, and spans weeks to months: a demanding scope, but one with considerable potential value.

The double meaning in this document’s title is deliberate. “Erasing the pain” refers to the patient’s pain: the persistent, disabling, expensive burden of low back pain. But it also refers to the system’s pain: the friction, duplication, heterogeneity and opacity that makes managing this condition through current healthcare models so chaotic. This project wants to address both.

The fellowship question and scope

The framing question driving this work is simple: can we build an end-to-end system that supports the clinical management of low back pain using openEHR for clinical knowledge representation and data persistence, while effectively orchestrating the clinical process using available technologies?

To answer it, the fellowship pursues three interlocking lines of work. The first is reaching clinical consensus on a minimal, implementable LBP pathway, not a comprehensive guideline review, but a decision about what a team of clinicians can agree on as a computable starting point. The second is modelling that pathway using the process representation tools available in the openEHR and broader clinical informatics ecosystem (GDL2, Task Planning, and BPMN) to understand what each can and cannot express. The third is evaluating how a process engine might execute those representations in practice, and what that reveals about readiness of the tooling.

The document that follows is structured as a decision log as much as a technical report. Each section surfaces trade-offs, makes choices explicit, and leaves breadcrumbs for those who follow. It does not aim to settle the question of how clinical process support should be built; it aims to navigate it honestly, through a real pathway, with real tools, and with the friction that entails.

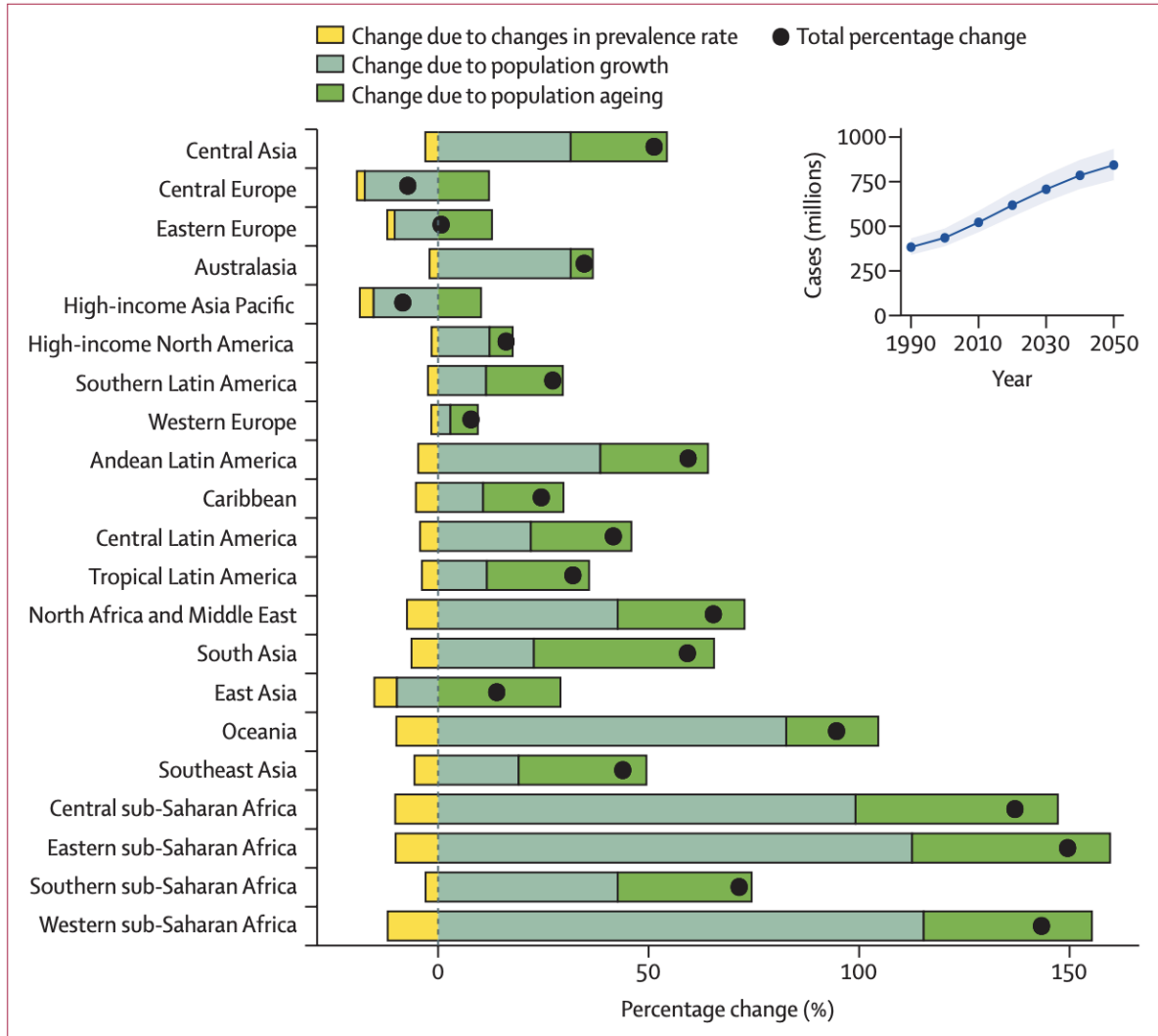


Figure 2: Decomposition of projected change in the number of prevalent low back pain cases between 2020 and 2050. Extracted from ([Ferreira et al. 2023](#))

The Clinical Context: Low Back Pain as a Process Problem

Understanding why low back pain was chosen as this fellowship's test case requires more than epidemiological justification. The condition occupies a particular clinical niche: common enough to represent a systemic challenge, structured enough to be modelable, and heterogeneous enough to make it genuinely difficult to standardise. This section provides the clinical grounding necessary to follow the modelling decisions that occupy the rest of the document. Not a guideline review, but the minimum context needed to understand why the pathway is shaped the way it is.

Burden and system pressure

Low back pain is the leading driver of years lived with disability in Spain and one of the highest-volume reasons for consultation across every level of the healthcare system. The estimated annual cost attributed to low back pain in Spain reaches €8,946 million, equivalent to 0.68% of GDP, with indirect costs from absenteeism and presenteeism accounting for nearly three-quarters of that figure. In Catalonia, approximately 17.2% of the population over fifteen reports current low back pain; chronic low back pain affects roughly one in five Spanish adults. The condition generated 393,575 temporary disability processes in Spain in 2015 alone, making it the leading cause of short-term work incapacity ([Alonso-García and Sarría-Santamera 2020](#); [Ministerio de Sanidad 2018](#)).

The downstream pressure on specialist services is equally striking. As of December 2025, 93,072 patients in Catalonia were waiting for a first orthopedics consultation, with a mean wait of 188 days, a significant share of that backlog driven by musculoskeletal and spinal pathology. Yet the primary care system absorbs the vast majority of low back pain episodes, and many of them could be fully managed at that level if the right clinical support were in place. The problem is not the absence of guidelines; it is the absence of tooling that embeds those guidelines into the care workflow.

A paradigm shift that reshapes the pathway

The clinical management of low back pain has undergone a fundamental reorientation over the past two decades. The traditional biomedical model, organised around identifying a

structural lesion as the explanation for pain, has proved insufficient for the majority of cases. Imaging findings correlate poorly with symptoms; many structural abnormalities are present in entirely asymptomatic individuals. Systematic use of imaging in the absence of alarm signals increases patient anxiety, generates over-diagnosis and drives avoidable interventions. Current international guidelines converge on reserving diagnostic imaging for presentations where the result would materially change the clinical decision ([Brinjikji et al. 2015](#)).

The contemporary framework is biopsychosocial. It recognises that biological, psychological and social factors interact to determine pain intensity, functional limitation and risk of chronification. Beliefs about pain, fear of movement (kinesiophobia), catastrophising, anxiety, depression and occupational context have been shown to influence clinical trajectories at least as strongly as structural findings. Approximately 39% of patients with chronic pain present clinically significant depressive symptoms; in low back pain specifically, depressive symptoms are associated with greater disability, higher healthcare consumption and poorer therapeutic response ([Aaron et al. 2025](#)).

This paradigm shift has direct implications for process modelling. A pathway structured around biomedical diagnosis leads to a relatively linear decision tree. A biopsychosocial pathway requires stratification, multi-axis evaluation and adaptive routing, the kind of conditional logic that benefits from explicit, computable representation rather than a PDF flowchart.

The clinical circuit: structure without rigidity

The model adopted in this fellowship draws on the care pathway developed by ICS (Institut Català de la Salut) in its consensus group called PADEICS, an integrated, transversal approach aligned with NICE and broader European consensus, and treats primary care as the axis of the entire circuit ([Institut Català de la Salut 2024](#)).

At first contact, the clinician’s task is not to establish an anatomical diagnosis but to perform a structured triage: exclude alarm signals (*red flags*), identify the predominant clinical phenotype, assess pain intensity and functional impact, and initiate patient education alongside active management. Five phenotypes organise the initial classification: *non-specific axial pain*, *non-radicular referred pain*, *radicular pain*, *lumbar canal stenosis* and *suspected specific pathology (inflammatory, infectious, neoplastic or systemic)*. Each phenotype carries a different default pathway and escalation logic, and the presence of red flags triggers an entirely separate urgent circuit bypassing the standard flow.

A planned early reassessment at two to six weeks constitutes the most consequential decision node in the pathway. Patients with favourable evolution continue in primary care. Those with persistent or worsening symptoms, significant functional limitation, psychosocial risk factors, or poor response to initial management are formally stratified using the STarT Back Screening Tool, an instrument that classifies patients into low, medium or high chronification risk and maps each stratum to a specific care configuration. Low-risk patients remain in primary care with reinforced self-management; medium-risk patients are referred early to physiotherapy

in primary care (FisioAPiC) with structured exercise and pain education; high-risk patients require a biopsychosocial approach involving REBEC (psychologists in primary care), and potentially Rehabilitation, Mental Health, the Pain Unit or a multidisciplinary spinal surgery team depending on clinical complexity.

For high-risk or persistently complex cases, additional assessment tools may be applied: PHQ-9 and GAD-7 for emotional symptoms, the Pain Catastrophizing Scale (PCS) and the Tampa Scale of Kinesiophobia (TSK) for psychological amplifiers. These are not proposed as universal screeners but as targeted instruments for patients whose trajectory suggests the standard care bundle will be insufficient.

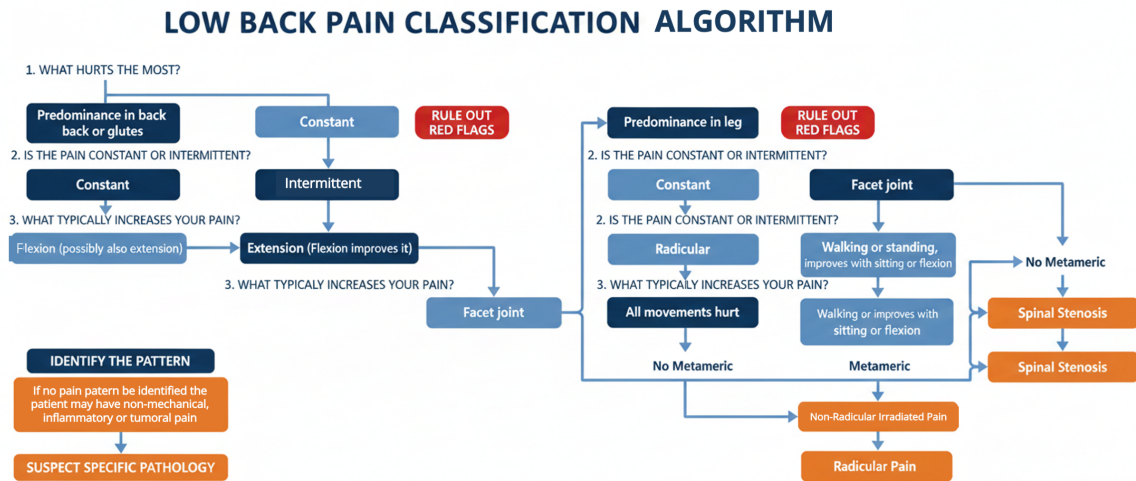


Figure 1: Algorithm proposed by PADEICS for low back pain classification (Modified from [Institut Català de la Salut 2024](#))

Variability as the modelling challenge

What makes this pathway both realistic and technically interesting is not the complexity of any individual branch, but the conditional logic that emerges from the interaction of clinical phenotype, risk stratum, timeline, patient response and local resource availability. The same presenting condition can legitimately route to entirely different care configurations depending on findings at the two-to-six-week node. Without explicit, shared decision rules, this variability resolves differently depending on the clinician's habits, the resources available in a given

territory, and the degree to which the record system surfaces the relevant decision criteria at the right moment.

This is precisely the problem that computable pathway representation is meant to address. The goal is not to constrain clinical judgement or flatten legitimate variation; it is to make the intended logic visible, traversable and auditable, so that the pathway does what it says it does consistently across providers and settings.

Key takeaways

Low back pain is a high-burden condition whose clinical management has shifted fundamentally from a biomedical to a biopsychosocial paradigm, with direct consequences for how the pathway must be structured. The care circuit is primary-care-led, phenotype-informed at triage, and risk-stratified at the critical follow-up node using the STarT Back tool. Each risk stratum maps to a defined care configuration, and high-complexity patients require an escalating, interdisciplinary response that spans multiple services and timescales. It is precisely this combination of structure and conditional heterogeneity that makes low back pain a demanding but appropriate test case for computable process support, and it is this clinical logic that the modelling work in subsequent sections will attempt to represent faithfully.

The Process Modelling Landscape

Anyone who has tried to move a clinical pathway from a PDF flowchart into a computable form has felt the mismatch: the tools available were mostly designed for parcel delivery, insurance claims, and factory floors. This section surveys the technical landscape of clinical process modelling: what is available, what each approach offers, and where each falls short when confronted with the messy reality of clinical care.

Why clinical process modelling is not generic BPM

The distinction matters. Industrial workflow automation assumes a largely deterministic environment: a parcel either arrives or it doesn't; a form is either complete or it isn't; the next step follows predictably from the previous. Healthcare process automation faces an adversary that these approaches were never designed for, the non-deterministic patient. A patient may not turn up for their appointment. The MRI results may not have arrived. A medication may produce an unexpected reaction. The clinician may override the recommended decision path based on contextual information that no model captures.

This has deep implications for tooling. A process formalism that handles exception paths awkwardly, or that multiplies Activity boxes every time a human needs to check data from an external system, becomes unworkable at clinical scale. The gap between the industrial workflow world and the clinical one is not merely a matter of domain vocabulary, it is a matter of what the formalism was fundamentally designed to do.

The OMG suite: BPMN, CMMN, and DMN

The most widely deployed clinical process modelling tools come from the Object Management Group (OMG). BPMN (Business Process Model and Notation) is the dominant language for structured, sequential processes. Its strengths are genuine: graphical clarity, a rich execution ecosystem via engines such as Camunda and Flowable, and broad adoption that spans both technical and clinical audiences. In health, it has found a natural home in care pathway visualisation, referral circuits, and hospital logistics. The BPM+ Health initiative, first at HIMSS and the US Veterans Health Administration and more recently moved to HL7, has invested considerable effort in extending BPMN's applicability to clinical settings ([Health Level Seven International 2026](#)).

Where BPMN shows its limits is precisely where clinical care departs from the deterministic. Exception handling must be manually encoded in the main workflow; every wait state, every external system interaction, becomes an additional Activity box. High-variability or event-driven clinical scenarios, such as the long-term care plan for a patient with multiple comorbidities or the rehabilitation programme that adapts to pain response, do not sit naturally inside BPMN's sequential logic.

CMMN (Case Management Model and Notation) was developed as a companion standard for exactly these adaptive scenarios, allowing tasks to be activated by events or conditions rather than ordered sequence. And DMN (Decision Model and Notation) handles the decision layer: rule tables and logic trees that encode clinical criteria, derivation thresholds, or alert conditions in an auditable form. In principle, BPMN+CMMN+DMN constitutes a complete process representation suite. In practice, the three are separately designed, use different meta-models, and require integration effort that multiplies with pathway complexity.

The FHIR layer: PlanDefinition and CQL

CQL (Clinical Quality Language), developed by HL7, approaches the process space from the opposite direction: it starts from FHIR resources and provides a formal language for expressing clinical rules, quality indicators, and decision logic directly over those resources ([Health Level Seven International \(HL7\) 2025a](#)). It integrates naturally with FHIR's Clinical Reasoning Module and with CDS Hooks, making it the standard choice for decision support in FHIR-native environments. **PlanDefinition** is FHIR's resource for representing sharable, computable clinical protocols and decision logic (order sets, care pathways, clinical decision support rules, or screening protocols) as machine-executable artifacts ([Health Level Seven International \(HL7\) 2025b](#)). Its limitation is its scope: CQL operates within the FHIR paradigm and does not integrate directly with openEHR data models. For environments where data is persisted in a Clinical Data Repository using openEHR archetypes, CQL's applicability is indirect at best. Additionally, you cannot express a high level of granularity with **PlanDefinition** and/or other FHIR artifacts.

The openEHR approach: GDL2 and Task Planning

The openEHR ecosystem contains two native approaches to process and decision representation. GDL2 (Guideline Definition Language, version 2) was designed to express clinical rules and guideline-derived logic as computable artefacts bound directly to openEHR archetypes ([openEHR International 2024a](#)). Its semantic alignment with the openEHR data model is a genuine strength, as rules operate on the same information structures used for data persistence, preserving meaning throughout. As of 2026, however, GDL2 is only available through an engine provided by Cambio group.

openEHR Task Planning (TP) was developed from 2015 onward specifically to address the limitations of industrial workflow tools in clinical settings ([openEHR International 2024b](#)). Its design principles are instructive. Rather than adapting a logistics formalism to healthcare, Task Planning was conceived around adaptive workflow: the user can skip tasks, repeat steps, abandon a plan, or switch to an exception path, all of which reflect clinical reality. Interaction with external systems (EMRs, order entry, CDRs) is built into each Task rather than represented as additional Activity boxes, keeping workflow definitions comprehensible. Exception handling is native, not manually encoded. A Decision Logic Module (DLM) handles decision representation within the same formalism, avoiding the fragmentation of BPMN+DMN.

The challenge is maturity. Task Planning remains experimental with few active implementations, and its tooling support is significantly thinner than the BPMN ecosystem. The specification exists; production-grade execution engines largely do not.

Comparison

	BPMN (+CMMN, +DMN)	CQL / FHIR CDS	GDL2	openEHR Task Planning
Primary domain	Industrial/logistics	FHIR-native	openEHR-native	openEHR-native
Execution tooling	Mature (Camunda, Flowable...)	Moderate	Limited	Experimental
Non-determinism	Weak (CMMN supplements)	Not addressed	Not addressed	Strong (by design)
Exception handling	Manual encoding	Not addressed	Not addressed	Native
Decision integration	Separate DMN	CQL	GDL2 logic	Integrated DLM
openEHR data binding	Indirect	None	Direct	Direct
Maintenance status	Active	Active	Active	Experimental

Recurring challenges

Across all these approaches, several structural challenges recur. **Terminology binding**, ensuring that a process step referencing “physiotherapy referral” connects to the same coded concept in every system it crosses, is not solved by process modelling alone; it requires explicit

governance of the terminology layer. **Decoupling clinical knowledge from process logic** remains an unsolved tension: guidelines change, and if decision logic is baked into workflow definitions rather than maintained as versioned artefacts, updating a guideline means reopening every workflow that embeds it. **Versioning** of both process models and their underlying data models is a governance problem as much as a technical one. And **interoperability**, the ability to exchange not just data but executable clinical knowledge across systems, remains largely aspirational in a landscape where each approach assumes its own runtime environment.

These are not arguments against formalisation. They are arguments for making these trade-offs explicit before choosing a toolset, which is what the next section attempts to do.

Key takeaways. No single process modelling standard satisfies all the requirements of clinical workflow in its current state. The OMG suite (BPMN/CMMN/DMN) offers the most mature tooling but at the cost of a fragmented formalism and design assumptions borrowed from industrial engineering. CQL is well-suited to FHIR-native decision support but does not reach across the openEHR boundary. openEHR Task Planning is arguably the most conceptually appropriate tool for clinical adaptive workflow, but its experimental status and thin implementation ecosystem impose serious practical constraints. The choice between them is not primarily a question of technical correctness; it is a question of what is buildable, maintainable, and executable given the constraints of the project at hand.

Low Back Pain Process Modelling Decisions

Taking the LBP pathway from a clinical guideline to a computable process model requires a series of concrete decisions that the literature rarely makes explicit: how to decompose the pathway into manageable units, how to handle decision logic, and how to connect the process layer to the data layer without conflating the two. This section walks through those decisions, the reasoning and the compromises behind each.

Choosing BPMN over Task Planning

The previous section established that openEHR Task Planning (TP) is arguably the most conceptually appropriate tool for clinical workflow. Yet the LBP model was built in BPMN, executed on Flowable. The reason is straightforward, if not fully satisfying: toolchain maturity won.

At the time of building, TP had no production-grade execution engine and no established integration pattern for connecting to a CDR. BPMN, by contrast, has a rich ecosystem. Flowable provided an executable engine, a visual designer, a REST API for process management, and out-of-the-box support for external workers and embedded DMN evaluation, all of which the PoC required. The trade-off accepted was real: BPMN's sequential assumptions push complexity back into the model for anything adaptive or exception-driven, and exception handling that TP would absorb natively must be explicitly encoded in BPMN's process flow. For this PoC, with a well-defined pathway, that trade-off was acceptable. For a more adaptive chronic care scenario, it would warrant re-evaluation.

The initial visualisation of the pathway was, in fact, done in TP's visual language, which made the clinical structure immediately legible. Migrating that representation to BPMN was not a translation without loss. Several TP concepts, among them task suspension, conditional re-entry, and the discipline of separating work planning from work execution, do not have clean BPMN equivalents. What resulted was a BPMN model that holds the structure of the pathway but encodes adaptability as explicit conditional branches rather than as a property of the tasks themselves.

Decomposing the pathway: call activities as clinical phases

All BPMN and DMN models can be found on GitHub ([Michelena 2026](#)). While the content has been translated into English, titles and identifiers have been intentionally preserved in

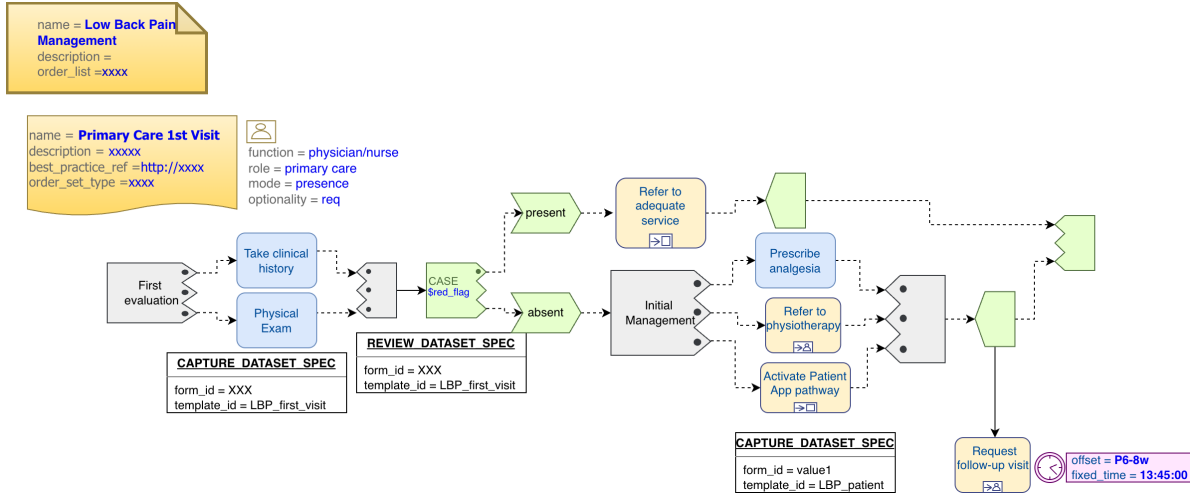


Figure 1: First draft of the initial phase of the Low Back Pain process modelled in Task Planning Visual Language

their original language to prevent broken references. Key terms for orientation: Lumbalgia (low back pain), Primaria (primary care), Diagnostico (diagnosis)...

The main process (**Lumbalgia_Primaria_Main**) is intentionally thin. It handles process initialisation, the flag-driven triage decision, and process finalisation. The clinical content, covering anamnesis, diagnosis, treatment, and follow-up, lives in four separate sub-processes invoked as call activities.

A call activity in BPMN is a reference to another independently deployable process definition. When the main process reaches a call activity, it instantiates the child process, waits for it to complete, and then resumes. This is not merely a structural convenience: it allows each clinical phase to be maintained, versioned, and eventually replaced independently of the others. It also allows explicit parameter mapping: the main process passes **ehrId**, the administrative composition identifier, and the process definition reference into each sub-process, giving each phase the context it needs to interact with the CDR without inheriting the entire variable scope of the main process.

The decomposition into Anamnesis, Diagnosis, Treatment, and Follow-up was not derived from the BPMN formalism; it was derived from the way the clinical team conceptualised the pathway. BPMN's call activity mechanism allowed that clinical decomposition to be preserved in the executable model, which is not always the case when process modelling forces a flat structure for execution reasons.

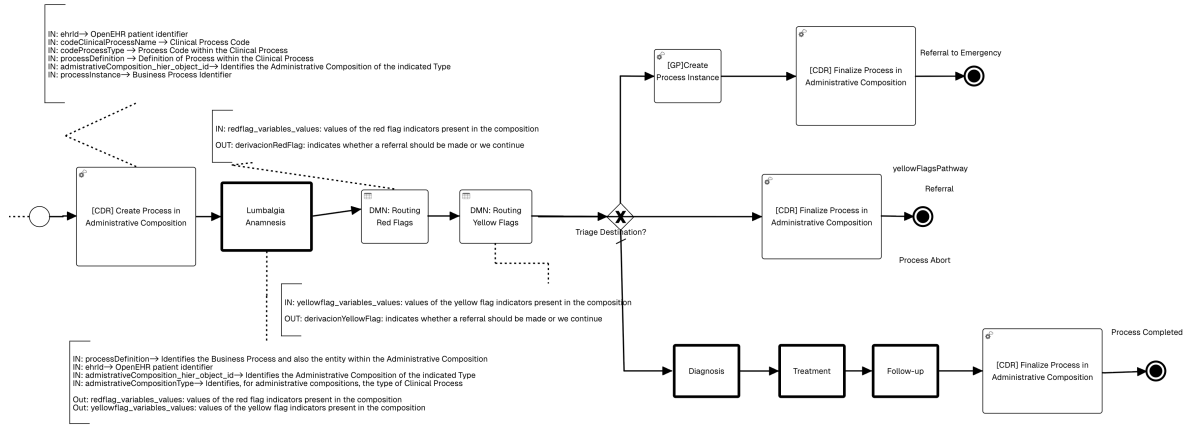


Figure 2: Simplified version of the Lumbalgia_Primaria_Main BPMN model

External workers and CDR integration

Every interaction with the Clinical Data Repository is modelled as an external worker task (`flowable:type="external-worker"`). An external worker is a service task whose execution is not handled internally by the process engine. Instead, the engine publishes a job to a topic queue; an external microservice subscribes to that topic, executes the work (in this case, calling the CDR's openEHR API) and returns the result, which the engine then maps back into the process variable space.

This pattern is deliberate. It separates the process orchestration concern (Flowable's responsibility) from the CDR interaction concern (the integration layer's responsibility) at the execution boundary. The BPMN model specifies what data flows in and what is expected out; the implementation detail of how to compose an openEHR AQL query or POST a COMPOSITION lives entirely outside the process definition. Changing the CDR endpoint, adding retry logic, or switching CDR vendors does not require touching the BPMN model, only the microservice implementation.

In practice, the external worker pattern introduces latency and operational complexity. Each CDR interaction becomes an asynchronous exchange through a job queue, which adds infrastructure to manage. For a PoC this overhead is negligible; for a production deployment serving hundreds of concurrent patient episodes, the job queue design and dead-letter handling become serious engineering concerns.

The parameters passed into and out of each CDR external worker are explicitly typed in the BPMN definition via `<flowable:externalWorkerInParameter>` elements. This makes the data contract between the process model and the CDR integration layer visible and auditable, a significant advantage over implicit parameter passing.

Decision logic via DMN

The two clinical branching points in the pathway, whether to start the LBP process at all and where to route the patient after flag assessment, are encoded as DMN decision tables embedded within the BPMN process, not as gateway conditions in the flow itself. This is defined by some as a separation of what-to-do-next (process logic) from why-to-do-it (clinical rule).

The first DMN (**reason_for_encounter**) acts as the process entry gate. It takes the diagnostic code from the Reason for Encounter field in the generic VISIT template and returns a boolean (**startProcess**) together with process type metadata.

The second DMN (**lumbalgia_flags_routing**) evaluates two boolean inputs, **redFlagsPresent** and **yellowFlagsPresent**, and returns a **triageDestination** string with three possible values: **urgencies**, **yellowFlagsPathway**, or **continue**. The hit policy is FIRST: if red flags are present, the rule fires immediately and routes to emergency referral regardless of yellow flag status. This captures the clinical priority ordering without requiring nested conditions in the process flow.

Encoding these rules as versioned DMN artefacts, rather than as hardcoded gateway conditions, has a direct governance benefit: when the clinical guideline changes, as LBP guidelines regularly do, the DMN table can be updated and redeployed without touching the process model.

Use of corticosteroids or immunosuppressants (rf_corticosteroid_immuno) <i>Boolean</i>	Parenteral drug addiction (rf_drug_iv) <i>Boolean</i>	High-risk age (<20 or >50-55 years) (rf_age_alert) <i>Boolean</i>	Red Flags Referral Destination (redFlagDestination) <i>String</i>
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Figure 3: Extract (incomplete) of the decision table that rules de DMN: Routing Red Flags

Exclusive gateways and triage branching

Following DMN evaluation, an exclusive gateway (**gw_triageDestination**) directs the process to one of three paths. The gateway is exclusive in the strict BPMN sense: exactly one outgoing path fires, determined by evaluating the conditions against the process variables set by the DMN. In this model, the conditions are simple boolean tests on **derivacionRedFlag** and

`derivacionYellowFlag`; the default path, taken when neither flag is positive, leads into the standard Diagnosis → Treatment → Follow-up sequence.

The yellow flags branch terminates at a placeholder end event. This is not an oversight; it reflects a genuine gap in the clinical specification at the time of modelling. The ICS guideline acknowledges that yellow flag management typically involves psychosocial intervention, pain unit referral, or active physiotherapy with a cognitive component, but the specific routing protocol had not been agreed with the clinical team. Rather than encoding an assumption, the model uses a labelled termination event (**Process Abort**) that makes the placeholder explicit.

Key takeaways

The LBP BPMN model represents a set of principled compromises more than an optimal solution. BPMN was chosen for its execution maturity, accepting that its sequential assumptions would require explicit encoding of what Task Planning would handle natively. Decomposition into call activities preserved clinical phase boundaries in an independently maintainable structure. External worker tasks gave the CDR integration layer a clean boundary and made the data contract auditable. DMN tables removed clinical decision logic from the process flow and positioned it as independently versioned artefacts aligned with guideline governance. Where the clinical specification was incomplete, most notably the yellow flags pathway, the model made that incompleteness visible.

Health Data Modelling Decisions

The process model gives the clinical pathway its shape; the data model gives it its substance or the subject proxy representation as Thomas Beale outlines in Task Planning. This section explains how the clinical content was structured in openEHR templates, how that structure connects to the process layer described in the previous section, and where the non-trivial decisions in data modelling lay, including one that required a workaround to bridge a gap in the CDR implementation.

Three templates for one pathway

These are the templates used in this process (please see github for more details ([Michelena 2026](#))):

- `Visit_generic_v0.2`: the generic outpatient visit (`COMPOSITION.encounter.v1`), maintained by the CatSalut clinical knowledge modelling team
- `Lumbalgia_Checklist_CD_v0.3`: the LBP-specific red and yellow flags checklist (`COMPOSITION.report.v1`)
- `POC_Care_Process_Orchestration_Record_v0.10`: an administrative composition bridging the process layer and the CDR (`COMPOSITION.process_list.v0`)

The generic visit template was not designed for this project. It is the institutional standard for outpatient encounters across CatSalut, assembling archetypes for reason for encounter (`EVALUATION.reason_for_encounter.v1`), narrative anamnesis (`OBSERVATION.story.v1`), physical examination (`OBSERVATION.exam.v1`), diagnosis (`EVALUATION.problem_diagnosis.v1`), clinical plan (`EVALUATION.plan_note.v0`), prescriptions (`INSTRUCTION.medication_order.v3`), and service requests (`INSTRUCTION.service_request.v1`). Reusing it here was a deliberate alignment with the institutional modelling strategy: clinical process support should draw on the same openEHR artefacts used across the system, not proliferate pathway-specific variants of generic encounter structures. The LBP pathway adds specificity through the checklist template and through the process layer, not by cloning and customising the encounter template.

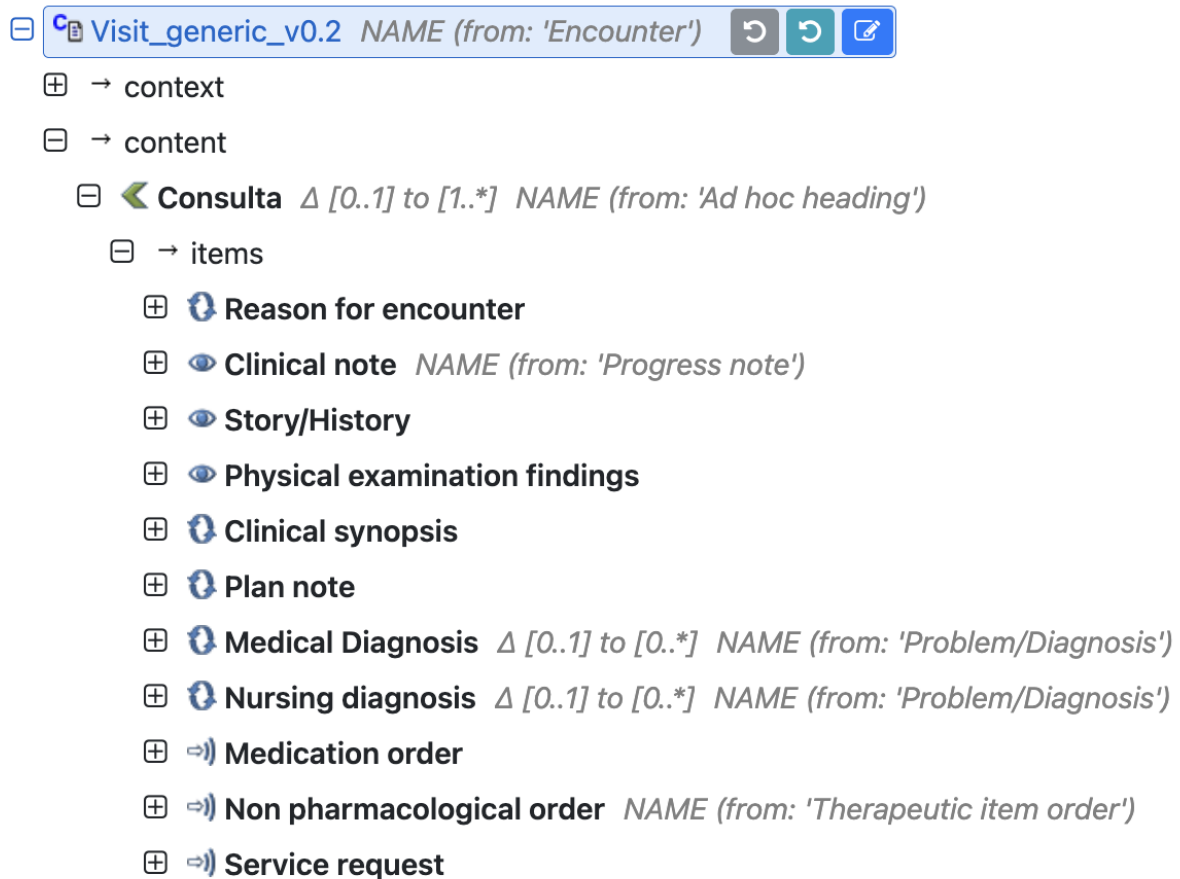


Figure 1: Main structure of the Visit_generic_v0.2 template

How the template structure supports the process model

Each of the three process triggers described in the previous section draws its data from a specific archetype within a specific template.

The first trigger lives in the `Visit_generic_v0.2` template. The `EVALUATION.reason_for_encounter.v1` archetype holds the diagnostic code that DMN1 evaluates to decide whether to start the LBP clinical pathway. The process engine does not read this directly: an external worker issues an AQL query to the CDR after the visit COMPOSITION is committed, extracts the coded value into the `reasonForEncounterCode` process variable, and passes it forward. The template is therefore both a clinical record and a trigger container, a dual role that works as long as the AQL extraction contract is maintained and the terminology binding in the visit template is consistent with the code set evaluated in DMN1.

The second and third triggers live in the `Lumbalgia_Checklist_CD_v0.3` template. Each red flag is recorded as an instance of `OBSERVATION.symptom_sign_screening.v1`; the age-threshold red flag uses `OBSERVATION.age_assertion.v1` as a separate archetype in the same composition. After the clinician completes and commits the checklist, a CDR external worker extracts boolean summaries into `redFlagsPresent` and `yellowFlagsPresent`, which feed the two DMN flag-routing tasks in the Anamnesis sub-process.

The red flags archetype decision

Modelling red flag checklists in openEHR proved less straightforward than anticipated, and the decision taken here reflects a broader unresolved question that surfaces in the community when clinical guidelines mix screening logic with alert generation.

The core question was temporal framing: is a red flag assessment a persistent risk profile, or a point-in-time screening event? The answer determines which openEHR reference model class is appropriate. A risk assessment (`EVALUATION.risk_assessment`) models an ongoing, evolving clinical state, like the cardiovascular risk estimate updated over years. Red flag screening for LBP is clinically different: it is episodic, repeatable, and time-bound to the encounter. Re-screening at a subsequent visit should produce a new record, not update the previous one. This points unambiguously to `OBSERVATION`, whose entries are immutable points in time.

Within the `OBSERVATION` family, `OBSERVATION.symptom_sign_screening.v1` provides a framework for recording structured answers to pre-defined screening questions about symptoms, signs, or potential underlying conditions, which matches the purpose of a red flag checklist precisely. The outlier is the age criterion: a patient's age is neither a symptom nor a sign in the clinical sense, which is why `OBSERVATION.age_assertion.v1` was introduced as a separate archetype within the same composition rather than forcing age into the symptom screening frame.

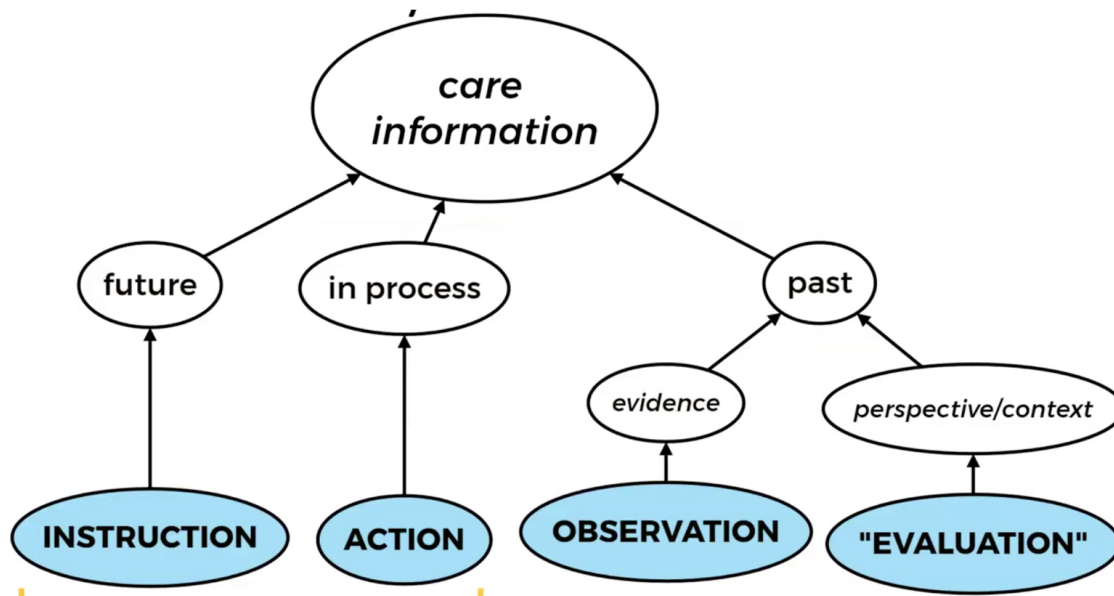


Figure 2: Conceptual algorithm to help clinical modelers decide which type of archetype is best for each care situation. Provided by the author, Heather Leslie.

The administrative composition: bridging process and data

One architectural gap encountered early in the PoC was the absence of a reliable mechanism in EHRbase to store a process engine instance identifier inside the EHR folder structure. The intent had been to record the business process instance identifier, the ID assigned by Flowable to each running pathway, at the folder level, so that every composition belonging to a given clinical episode could be linked back to its corresponding process instance.

EHRbase's folder implementation at the time did not support arbitrary metadata on folder nodes. The workaround was the third template: the `POC_Care_Process_Orchestration_Record`, an administrative `COMPOSITION` built on `COMPOSITION.process_list.v0`. This composition records the process instance identifier, the clinical process code, the process type, and the process lifecycle state as it changes over the course of the pathway. External worker tasks in the BPMN model write to this composition at key lifecycle transitions: process start, sub-process phase entry, and process finalisation or abort. The lifecycle state is drawn from a controlled vocabulary with three terminal values: **Finalized** (pathway completed normally), **Finalized - referred to emergency** (red flags triggered early exit), and **Aborted - yellow flags pathway** (pending pathway, process suspended).

The cost of this approach is a composition in the patient's EHR that carries no clinical content, only process metadata. The benefit is that it enables structured AQL queries over process

state: which patients are currently in the Anamnesis phase, which episodes ended with an emergency referral, how many processes were aborted during the PoC. In the absence of folder-level metadata support, the administrative composition is the only reliable record of process provenance within the CDR.

Folder structure and episode classification

The folder structure deployed in the PoC follows the two-axis model used more broadly in CatSalut's openEHR implementation. The first axis is administrative, grouping compositions by institutional provenance (the originating centre). The second axis is clinical, grouping them by care pathway or episode, in this case the LBP folder under which all compositions belonging to a given episode are referenced.

The key property of openEHR folder items is that they are references to `VERSIONED_COMPOSITION` objects, not copies. The same composition can appear under both the administrative axis (GP Sants) and the clinical axis (Low back pain) without duplication, enabling two independently useful views over the same data. This cross-axis referencing was one of the capabilities explored as part of the PoC, and it worked as specified, though for folder-based AQL, EHRbase relies on an experimental extension rather than the standardised AQL specification. That extension is capable and well-documented, but it introduces a dependency on EHRbase-specific behaviour that limits portability to other openEHR CDR vendors.

Key takeaways

The data modelling decisions for the LBP PoC were shaped at every turn by the need to align with existing institutional artefacts, accommodate constraints in the CDR implementation, and resolve genuine openEHR specification gaps at the boundary between data persistence and process execution. The checklist template's choice of symptom screening `OBSERVATION`s over persistent `EVALUATION`s was a principled clinical argument, though the long-term answer is still open. The administrative composition was a pragmatic workaround for missing folder metadata support, adding process-layer traceability at the cost of a clinically empty record in the EHR, a debt that should be retired if EHRbase or an equivalent CDR gains native folder-level metadata. Together, these three templates define the data contract that the process model depends upon, and each carries a technical debt that future iterations of this architecture should aim to resolve.

Discussion

Building a computable clinical pathway is an exercise in discovering what you did not know you did not know. The LBP proof of concept achieved what it set out to do: a functional BPMN process model connected to an openEHR CDR, with executable decision logic, structured data templates, and a process-data bridge that makes the episode legible from both sides. That is a real result. But the experience of building it surfaces a set of structural questions that go well beyond any single pathway, questions about the right tools, the right abstractions, and the right division of labour between process and data layers that the field has not yet resolved.

Strengths of the approach

The combination of BPMN, DMN, and openEHR proved capable of representing the deterministic spine of a clinical pathway in a way that is both executable and auditable. Each technical layer stayed within its designated concern: the CDR persisted structured clinical data, the DMN tables encoded clinical decision rules as independently versioned artefacts, and the BPMN model orchestrated the sequence of clinical phases without embedding either data structures or rule logic directly. The external worker pattern, in particular, achieved a clean separation between process orchestration and CDR integration, a boundary that matters for maintainability and will matter even more for production deployment at scale.

The modular decomposition into call activities deserves special mention. Clinical stakeholders can inspect the Anamnesis, Diagnosis, Treatment, and Follow-up sub-processes as self-contained units, which maps to how clinicians conceptualise the pathway. The versioned DMN tables mean that when the clinical guideline changes, as it will, the update can be isolated to the decision artefact rather than requiring surgery on the process model. These are not small gains. Most clinical decision support systems conflate rule, data, and process in ways that make any one of them expensive to change without inadvertently disturbing the others.

The BPMN imperative and its cost

BPMN was chosen for execution maturity, and that pragmatism was correct. But the cost becomes visible the moment you try to model anything that deviates from the sequential deterministic path. The yellow flags branch, a clinically real and important fork in the pathway, sits as a placeholder in the current model precisely because BPMN provides no graceful mechanism for what is fundamentally an open task: acknowledge the psychosocial complexity,

send the patient into a different care context, and remain available to pick up the thread months later. Task planning in the clinical sense is not a sequence; it is a suspended set of intentions that can be reactivated, renegotiated, and revisited. BPMN, working from its origins in business process automation, makes that semantics expensive to represent.

The verbosity of the BPMN models themselves is a related symptom. What could be expressed in a few lines of Task Planning notation, a conditional task, a follow-up obligation or a care plan item, becomes a gateway, two intermediate boundary events, a collapsed sub-process, and a parameter mapping. Every piece of that verbosity is necessary in the execution engine; none of it is natural to a clinical reader.

Why the openEHR process stack deserves the next PoC

The honest reason for choosing BPMN over Task Planning was not a considered trade-off weighing the merits of both; it was the absence of a production-grade Task Planning execution engine at the time of building. That absence may be temporary. The theoretical case for openEHR's native process stack is strong, arguably stronger than the BPMN case, for several reasons that this project's experience makes concrete.

The first is the Decision Logic Module (DLM). The authors of the Task Planning specification made available some examples such as the `pregnancy_ruleset.dlm` which shows remarkable insights ([Beale 2021](#)). The DLM syntax can declare input variables with explicit semantic currency (`currency = 24h`), express risk stratification across multiple concurrent dimensions, and return the highest-risk aggregate of a set of clinical assessments, all in a form that remains recognisable to a domain expert. Compared to a DMN decision table, which encodes each rule as a row in a two-dimensional matrix, the DLM can represent the full epistemic picture of a guideline: what the variables mean, not just what values they take.

The second is the Subject Proxy. The openEHR Task Planning specification defines a Subject Proxy Service whose role is to act as a concentrator, drawing data from all available epistemic sources (CDR queries, lab systems, device feeds, and just-in-time clinician input) and lifting them into a stable, concrete representation of patient state for consumption by plans and decision logic ([openEHR International 2024b](#)). This is exactly the function that the external worker pattern approximates clumsily in the current PoC: AQL queries extracted into process variables, with no guarantee of recency, no temporal currency semantics, and no unified interface between the process model and all the systems that might hold relevant data. The Subject Proxy is not yet implemented in any production-grade form, but it makes a lot of sense to me.

The missing piece

Implementation guides are the missing connective tissue. This PoC required a series of decisions about how process instances map to EHR folder structure, how compositions record lifecycle

transitions, how terminology bindings in templates propagate into DMN trigger conditions, none of which are specified by the openEHR standard itself. These decisions should live somewhere. This represents, in my view, one of the openEHR community's outstanding challenges for achieving consistent cross-pathway and cross-organisational architecture: the structuring of episodes within the folder hierarchy, the role of master templates as data contracts for process triggers, and the conventions by which administrative compositions record provenance. Absent community-level consensus on these questions, architectural decisions are inevitably reinvented in isolation, and integration points remain fragile.

Key takeaways

This proof of concept demonstrated that BPMN plus DMN plus openEHR is a functional architecture for deterministic clinical pathway execution, capable, auditable, and maintainable where the pathway is well-specified. The external worker pattern and modular call activity decomposition are technical choices that held up well and would carry into production. But the verbosity of BPMN, its uncomfortable fit with adaptive clinical tasks, and the architectural friction of mapping openEHR data into process variables all point in the same direction: the openEHR native process stack, Task Planning, DLM, and the Subject Proxy Service, represents a more holistic and semantically richer vision for clinical process support than the BPMN ecosystem can offer. The gap is currently one of toolchain maturity, not of conceptual correctness. Closing that gap, through production implementations of the Task Planning execution engine and a Subject Proxy reference service, is one of the most challenging and ambitious work the openEHR community could make for clinical process support at scale.

Future Research Directions

The LBP pathway PoC in this fellowship has produced a working architecture, a set of concrete trade-offs, and a clearer picture of what the tooling cannot yet do. What it has not produced is a clinical process support system that a health organisation can deploy and depend on. Not yet.

Extending the approach to other pathways

The LBP pathway was chosen partly because it is relatively well-defined: a deterministic spine with clear branching conditions and documented guideline evidence. That made it a workable first target. The most important next test is whether the same architecture generalises to conditions with less deterministic structure: chronic conditions managed across multiple organisational levels, pathways where the guideline evidence is contested, or conditions such as diabetes management, heart failure monitoring and oncology follow-up, where the care model shifts between protocols at irregular intervals driven by patient response rather than calendar time.

Each of those extensions surfaces a different structural challenge. Multi-organisational pathways expose the CDR portability problem: AQL queries that work within a single EHRbase instance may not compose cleanly across a federated architecture. Contested guideline pathways expose the governance problem: who owns the DMN version, under what authority, and how does a clinician know which version a given process instance is running? Adaptive chronic care pathways expose the expressiveness problem that BPMN already struggles with in the LBP yellow flags case. Testing the architecture against these scenarios, with real clinical teams, is the most direct route to understanding whether the current design assumptions hold or need to be renegotiated.

A parallel priority is the yellow flags pathway that remains a placeholder in the current model. Completing that branch, in collaboration with the pain unit and physiotherapy teams that would actually own the care, is a tractable near-term goal with direct clinical value. It is also a test case for CMMN (Case Management Model and Notation) as a complement to BPMN for the adaptive, goal-oriented components of pathway management that do not fit naturally into sequential process flows.

The openEHR stack: what needs to mature

The discussion section identified two openEHR-native capabilities that this PoC had to work around rather than leverage: the Task Planning execution engine and the Subject Proxy Service. Both are specified at a level of theoretical clarity that makes their absence from production implementations genuinely frustrating. The research investment required here is not conceptual; the concepts are well-developed. It is an engineering challenge: reference implementations that health informatics teams can test against, benchmark, and integrate without building the infrastructure themselves.

GDL2 occupies an interesting position in this landscape. It exists and has been deployed in production (the Cambio engine remains the primary implementation), but it is not integrated with the process execution layer in the way that DLM is designed to be. Evolving GDL2, or defining a clear interoperability path between GDL2 guidelines and Task Planning DLM modules, would allow health systems that have already invested in GDL2-based decision support to participate in the process-aware architecture without discarding existing clinical knowledge artefacts. That evolution is worth pursuing.

The folder-level metadata gap that forced the administrative composition workaround in this PoC is a CDR maturity problem and probably an improvement needed in the specifications. EHRbase and equivalent CDRs should support the full blown specifications for folders, eliminating the need for a clinically empty administrative composition whose only purpose is to carry a process instance identifier. This is a concrete, bounded improvement request that the community could prioritise.

Deterministic process and the agentic AI horizon

The relationship between structured clinical process support and large language model-driven clinical reasoning is one of the more genuinely open questions that this work surfaces. The framing that a deterministic BPMN pathway and an LLM-driven clinical agent are alternatives, that AI will eventually replace the need for explicit process modelling, is, in my view, a category error. They operate on different uncertainties at different timescales.

The value of a deterministic process layer is not that it can make every clinical decision automatically; it is that it makes the care pathway legible, auditable, and governable. When a patient episode enters the red flags branch and triggers an emergency referral, the fact that this happened, and why, is recorded in a computable artefact that can be audited, counted, and compared against population norms. An LLM embedded in a clinical conversation may eventually produce a better narrative recommendation than a DMN table, but it does not, by design, produce a verifiable event log of clinical decisions against a guideline specification.

The more productive framing is complementary: deterministic BPMN or Task Planning processes as the governance skeleton, with LLMs and agentic reasoning filling in the contextual, conversational, and interpretive tasks that rule-based systems handle poorly. Tools like n8n,

which blend structured workflow automation with LLM integration nodes, are early experiments in this direction. They demonstrate that the combination is technically feasible and operationally useful, but they also reveal the problem clearly: when a node in an n8n workflow calls an LLM to interpret a clinical text and route a patient, the auditability of that routing decision is limited to the LLM's output token and whatever logging the workflow engine records. For clinical governance at a population level, that is not yet sufficient.

The research question is therefore not whether to use agentic AI in clinical process support, but where the boundary between deterministic and probabilistic reasoning should be drawn, how to make that boundary explicit in process models, and what audit semantics are required at the handoff. That question does not have a clean answer today, but it is the right question, and building the computable process layer described in this document is a prerequisite for asking it rigorously.

Key takeaways

The LBP pathway PoC points forward as clearly as it points backward. Extending the architecture to adaptive and multi-organisational pathways, completing the yellow flags branch with real clinical ownership, and validating the DLM against the current DMN implementation are near-term priorities that build directly on the existing foundation. Maturing the openEHR native process stack, covering Task Planning execution, Subject Proxy implementation, GDL2 integration, and folder-level metadata support in CDRs, is the structural investment that would make the architecture replicable at health-system scale rather than remaining a one-team PoC exercise. The relationship between deterministic process support and agentic AI is the most open question on the horizon, and it is one that only becomes tractable once the deterministic layer is robust enough to define where the boundary with probabilistic reasoning should sit. The pain of the work described in this document is, in the end, the pain of being early: the tools are ahead of their integrations, the specifications ahead of their implementations, the vision ahead of the infrastructure. Is the direction the correct?

References

- Aaron, R. V., S. G. Ravyts, N. D. Carnahan, et al. 2025. “Prevalence of Depression and Anxiety Among Adults with Chronic Pain: A Systematic Review and Meta-Analysis.” *JAMA Netw Open* 8 (3): e250268. <https://doi.org/10.1001/jamanetworkopen.2025.0268>.
- Alonso-García, M., and A. Sarriá-Santamera. 2020. “The Economic and Social Burden of Low Back Pain in Spain: A National Assessment of the Economic and Social Impact of Low Back Pain in Spain.” *Spine (Phila Pa 1976)* 45 (16): E1026–32. <https://doi.org/10.1097/BRS.0000000000003476>.
- Arikan, Seref. 2024. *What Is an EHR Slice and Why Is It Relevant for EHDS?* <https://www.linkedin.com/pulse/what-ehr-slice-why-relevant-ehds-seref-arikan-phd-mqphe/>.
- Beale, Thomas. 2021. *CDS, Guidelines and Planning Overview*. openEHR international. <https://specifications.openehr.org/releases/PROC/development/overview.html>.
- Brinjikji, W., P. H. Luetmer, B. Comstock, et al. 2015. “Systematic Literature Review of Imaging Features of Spinal Degeneration in Asymptomatic Populations.” *AJNR Am J Neuroradiol* 36 (4): 811–16. <https://doi.org/10.3174/ajnr.A4173>.
- Ferreira, Manuela L, Katie De Luca, Lydia M Haile, et al. 2023. “Global, Regional, and National Burden of Low Back Pain, 1990–2020, Its Attributable Risk Factors, and Projections to 2050: A Systematic Analysis of the Global Burden of Disease Study 2021.” *The Lancet Rheumatology* 5 (6): e316–29. [https://doi.org/10.1016/S2665-9913\(23\)00098-X](https://doi.org/10.1016/S2665-9913(23)00098-X).
- Health Level Seven International. 2026. *BPM+ Health*. <https://www.hl7.org/bpm/>.
- Health Level Seven International (HL7). 2025a. *Clinical Quality Language (CQL) Specification, v2.0.0-ballot*. <https://build.fhir.org/ig/HL7/cql/>.
- Health Level Seven International (HL7). 2025b. *PlanDefinition – FHIR v6.0.0-ballot4*. <https://build.fhir.org/plandefinition.html>.
- Institut Català de la Salut. 2024. *Guia de Pràctica Clínica Sobre Lumbalgia*. Institut Català de la Salut. <https://ics.gencat.cat/web/.content/Assistencia/esquena/guia-lumbalgia.pdf>.

Michelena, Xabier. 2026. *Advancing Towards Clinical Process Support: Deliverables of the openEHR Fellowship*. https://github.com/xabimich/openehr_fellowship_lbp; GitHub.

Ministerio de Sanidad. 2018. *Encuesta Nacional de Salud de España 2017*. Ministerio de Sanidad.

openEHR International. 2024a. *Guideline Definition Language v2 (GDL2)*. <https://specifications.openehr.org/releases/CDS/latest/GDL2.html>.

openEHR International. 2024b. *Task Planning (TP) Specification*. https://specifications.openehr.org/releases/PROC/latest/task_planning.html.